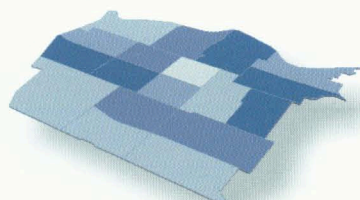


# Defining spatial neighborhoods and weights

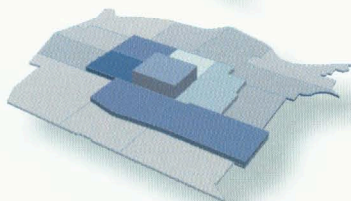
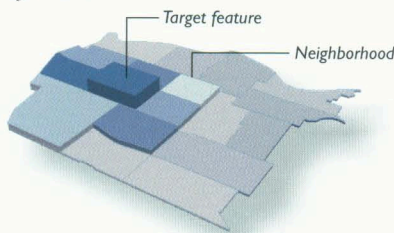
Several of the methods discussed in chapters 3 and 4 show you how to analyze patterns and clusters of feature values. These methods look at both the difference between the values of features and the spatial relationship between the features (distance or other measure).

Specifically, the GIS compares the value of a feature (the “target”) to the values of neighboring features. It then moves to the next feature and does the same thing, and so on, for all the features in the study area. In order to do this, the GIS requires that you define the area surrounding each target feature within which feature values are compared—termed the “neighborhood”—and the nature of the spatial relationship between features. The GIS then assigns weights to each feature pair to specify whether the two features are in each other’s neighborhoods, and to represent the spatial relationship between the features.

You define the neighborhood based on the interaction between features. Features might influence each other—for example, the value of a property is often strongly influenced by the values of neighboring properties. Or, there may be movement between features that causes the values of nearby features to be similar. For example, where there is an outbreak of a contagious disease in a particular census tract, nearby census tracts will also have high incidence of the disease as people move to and from the surrounding areas, infecting others. Or it might be that there are underlying factors resulting in nearby features having similar values. For example, two adjacent counties may have voted heavily for the same candidate because they are both in a farming area and the people living there share the same opinions as the candidate on issues related to agriculture.

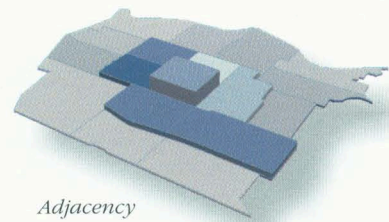


*Features color coded by value*

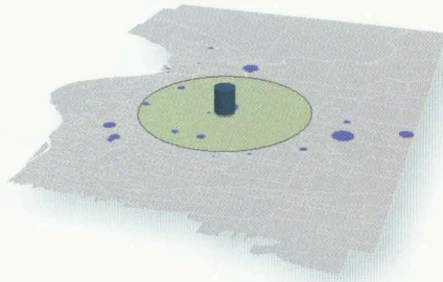


*Each feature in turn is compared to its neighbors.*

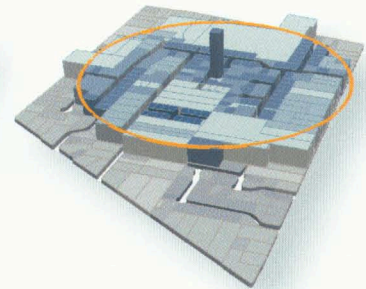
In most cases, these spatial relationships are represented using adjacency (contiguous features that share a border) or Euclidean distance. Adjacency-based neighborhoods are often used for contiguous areas or rasters, while distance-based neighborhoods can be used for discrete or contiguous features.



*Adjacency*



*Distance: discrete features*



*Distance: contiguous features*

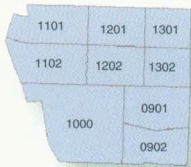
You could define a neighborhood using other measures, such as travel time, travel cost, measures of attraction between locations (such as the attraction of various resorts to cities), or any other measure of spatial relationships. However, these measures can be hard to obtain, or even to quantify.

The GIS implements a neighborhood using weight values. The GIS first determines the relationship between each pair of features in the study area—using the type of neighborhood you define—and assigns a weight value to that pair. The value is stored in a table (known as a spatial weights matrix) with a row and a column for each feature, populated by the weight for each feature pair. When the GIS calculates a particular statistic, the GIS looks up the weight in the matrix.

Some GIS software creates the matrix on the fly when you perform an analysis. It either uses a default neighborhood that is appropriate for the particular analysis method, or it prompts you for the type of neighborhood if there are several different neighborhood types that would be appropriate. Other GIS software allows you to create a weights matrix as a separate step before using a particular statistical tool.

For adjacency-based neighborhoods, a weight of 1 means the nearby feature is in the neighborhood. A weight of 0 indicates the feature is not adjacent and therefore not in the neighborhood.

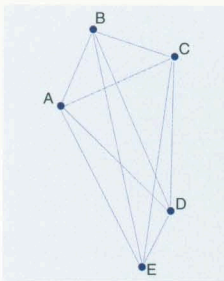




*Weights matrix for an adjacency-based neighborhood*

|      | 1101 | 1201 | 1301 | 1102 | 1202 | 1302 | 1000 | 0901 | 0902 |
|------|------|------|------|------|------|------|------|------|------|
| 1101 | 0    | 1    | 0    | 1    | 1    | 0    | 0    | 0    | 0    |
| 1201 | 1    | 0    | 1    | 1    | 1    | 1    | 0    | 0    | 0    |
| 1301 | 0    | 1    | 0    | 0    | 1    | 1    | 0    | 0    | 0    |
| 1102 | 1    | 1    | 0    | 0    | 1    | 0    | 1    | 0    | 0    |
| 1202 | 1    | 1    | 1    | 1    | 0    | 1    | 1    | 1    | 0    |
| 1302 | 0    | 1    | 1    | 0    | 1    | 0    | 0    | 1    | 0    |
| 1000 | 0    | 0    | 0    | 1    | 1    | 0    | 0    | 1    | 1    |
| 0901 | 0    | 0    | 0    | 0    | 1    | 1    | 1    | 0    | 1    |
| 0902 | 0    | 0    | 0    | 0    | 0    | 0    | 1    | 1    | 0    |

For distance-based neighborhoods, the weight value is calculated using the distance between the features, so each pair potentially has a unique weight value.

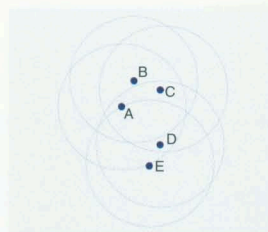


*Weights matrix for a distance-based neighborhood*

|   | A   | B    | C   | D   | E    |
|---|-----|------|-----|-----|------|
| A | 0   | 353  | 516 | 641 | 757  |
| B | 353 | 0    | 357 | 837 | 1025 |
| C | 516 | 357  | 0   | 659 | 901  |
| D | 641 | 837  | 659 | 0   | 263  |
| E | 757 | 1025 | 901 | 263 | 0    |

You can also create a neighborhood using a threshold or cutoff distance. Within this distance neighbors can have a weight of 1, or the weight can be calculated using the distance value.

*Weights matrices using a threshold distance*



|   | A | B | C | D | E |
|---|---|---|---|---|---|
| A | 0 | 1 | 1 | 1 | 0 |
| B | 1 | 0 | 1 | 0 | 0 |
| C | 1 | 1 | 0 | 1 | 0 |
| D | 1 | 0 | 1 | 0 | 1 |
| E | 0 | 0 | 0 | 0 | 0 |

|   | A   | B   | C   | D   | E   |
|---|-----|-----|-----|-----|-----|
| A | 0   | 353 | 516 | 641 | 0   |
| B | 353 | 0   | 357 | 0   | 0   |
| C | 516 | 357 | 0   | 659 | 0   |
| D | 641 | 0   | 659 | 0   | 263 |
| E | 0   | 0   | 0   | 263 | 0   |

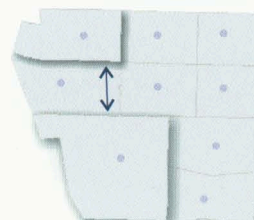
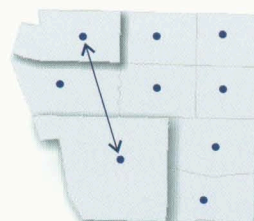
## IS THE INFLUENCE LOCAL OR REGIONAL?

If the influence of feature values on surrounding features is minimal or is limited to those close by, or the phenomenon is very localized, use a neighborhood based on adjacency. For example, if you're analyzing the vote for a particular candidate in an election by county (using that candidate's percentage of the vote as the attribute value), you'd use a neighborhood based on adjacent counties. In most places, the vote is localized, with urban counties voting differently than rural counties.

If distant features exert some influence on each other or have some interaction, even if minimal, you would use a neighborhood that includes all features in the study area, with decreasing influence as the distance increases. You'd use this type of neighborhood, for example, when analyzing the pattern of occurrence of a contagious disease. You can assume that people may travel (and infect others) throughout the study area, but that they are more likely to come into contact with those in nearby areas.

When you're using a distance-based neighborhood with point features, the GIS measures the point-to-point distance. For lines and areas you can measure the distance between feature centroids or between the nearest locations on the features.

- Distance between centroids. Use this distance if the features are roughly the same size and shape.
- Distance between nearest locations on line or boundary. Use this distance if the size or shape of the features varies a great deal—otherwise, the weight may be underestimated for particularly large or convoluted features since the centroid could be far from one end of the line or the border of the area.

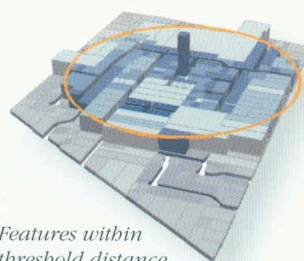


*The distance can be very different depending on whether you measure from centroid-to-centroid (top) or border-to-border (bottom).*



**IS THERE A DISTANCE  
BEYOND WHICH THERE  
IS NO INFLUENCE OR  
INTERACTION?**

If you know that the influence of features on each other drops off completely after a certain distance, you can define the neighborhood to reflect this. This is the case, for example, with property values, where there is an accepted rule of thumb that the influence of a property on others extends to about a quarter-mile—past that distance there is no longer much influence.

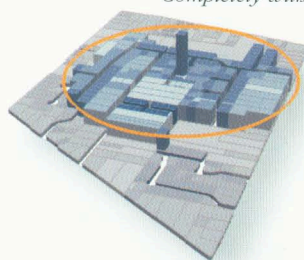


*Features within  
threshold distance*

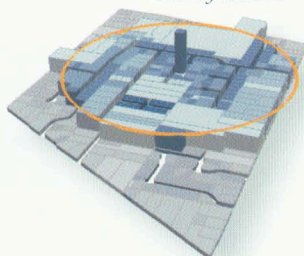
When using a cutoff distance with point features, a single point is either within the distance or not. For features other than points, however, including lines, areas, and rasters, the feature could fall partially outside the distance. You need to decide if a feature is included if it doesn't fall completely within the distance. The options are:

- only features that fall completely within the distance.
- features that fall mostly within the distance. This is usually defined as the centroid of the feature falling within the distance. In some cases features are included if half the length (for line features) or more than half the area (for area features) falls within the distance.
- features that fall partially within the distance—any part of the feature is within the distance.

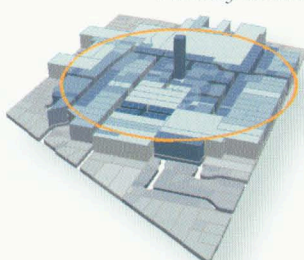
*Completely within*



*Mostly within*



*Partially within*

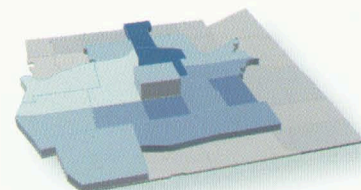
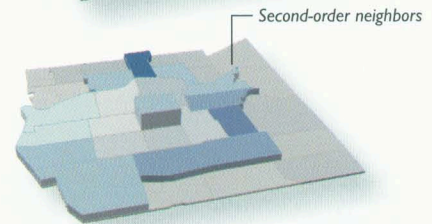
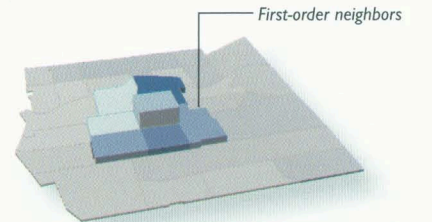


If there is interaction between the features, you might want to include features that fall partially or mostly within the distance, to account for that interaction. For example, if you're analyzing earthquake faults by mean magnitude to identify clusters of faults that have had earthquakes of similar strength, you'd include faults that fall mostly within the cutoff distance, since earthquakes on one fault can trigger quakes on nearby ones. You'd want to exclude faults that are mostly beyond the distance, since you can assume that most of the quakes on those faults occurred outside the distance.

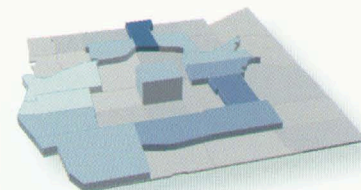
If there is no direct interaction, you might want to include only features falling completely within the distance. For example, when analyzing road segments by the average number of accidents per month to see if accidents form clusters, you'd exclude road segments falling partially within the distance, since—unlike with earthquakes occurring on nearby faults—the fact that one segment has a lot of events doesn't influence another segment to have a lot of events.

If you're using a neighborhood based on adjacency, you can include only those features immediately adjacent to each other (termed first-order features), or you can include features that are once-removed, twice-removed, or more. The latter are termed higher-order features. Second-order features would be those adjacent to the first-order features. Using higher-order features allows you to increase the extent of adjacency-based neighborhoods.

If you do use higher-order features, you can specify whether or not to include lower-order features in the neighborhood as well. Including lower-order features allows you look for patterns at different scales. Excluding lower-order features allows you to look at the dispersion of a geographic phenomenon over time—you could see if the pattern changes as you include neighbors at higher orders. If, for example, you're trying to determine how quickly an outbreak of flu is spreading, you'd run the analysis using first-order census tracts, then run it again using only second-order tracts, and so on. In this case, you're using the increasing orders as a surrogate for the passage of time.



*Neighborhood using first- and second-order features*



*Neighborhood using second-order features only*